

COMP3540 – HUMAN COMPUTER
INTERACTION AND DESIGN

TEAM PROJECT PHASE #5: EVALUATION

Team Members: Arshdeep Ghotra (T00684423), Nand
Bhasin (T00682801)

1. INTRODUCTION

FITBUDDY fitness tracker app project is a comprehensive digital tool designed to enhance users' physical and mental health. It offers features for tracking daily workout sessions, provides educational articles, workout templates, as well as daily water and nutrition tracking. Additionally, it includes a mental health section with resources for meditation and articles on mental wellness.

Current Status:

The core functionalities for workout tracking, educational content delivery, and nutritional guidance are in development.

The mental health section is partially implemented, with meditation resources available and plans to expand the articles provided.

Issues or Challenges:

Integrating a seamless user experience across diverse functionalities while maintaining app performance.

Curating and updating a vast library of educational content to ensure relevance and accuracy.

Developing a reliable system for tracking and analyzing nutritional intake and hydration levels.

Ensuring privacy and security of user data, especially with sensitive health information. the scope and features of your fitness tracker app project, establishing clear usability and UX (User Experience) goals is crucial for its success. These goals will guide the development process and inform usability testing to ensure the app meets user needs effectively.

2. Project Usability Goals:

Ease of Use:

Ensure users can navigate and utilize the app's features effortlessly, from tracking workouts to reading educational articles and using the mental health resources.

Accessibility:

The app should be usable by a wide audience, including those with disabilities, ensuring features like text-to-speech for articles and customizable UI elements for better visibility.

Reliability:

All features, from workout tracking to nutrition logging, must work correctly and consistently, providing accurate data and insights to users.

Performance Efficiency:

The app should load quickly and operate smoothly, even with extensive content or data, such as articles and workout templates.

Privacy and Security:

Secure users' personal and health-related data, ensuring trust and compliance with relevant regulations.

UX Goals:

Engagement:

Design an engaging and motivating user experience that encourages regular interaction with the app for workout tracking, educational content, and mental health resources.

Personalization:

Allow users to tailor their experiences, from custom workout plans to preferred content types, enhancing relevance and usefulness.

Learning and Discovery:

Facilitate easy discovery of new workouts, health tips, and mental wellness strategies, encouraging ongoing education and exploration.

Support and Guidance:

Provide supportive features like progress tracking, reminders for water intake and workouts, and tips to improve mental health, ensuring users feel guided and supported.

Focus for Usability Testing:

For the initial usability testing phase, the focus will be primarily on Ease of Use and Engagement:

Ease of Use:

It's critical to ensure that users find the app intuitive. This includes navigating between different sections (workouts, nutrition, mental health), accessing content, and logging activities. Usability testing will assess how easily new users can navigate and utilize the app without frustration.

Engagement:

Given the diverse features of the app, maintaining user engagement is essential. Testing will evaluate which features users interact with most, how they interact with the content (e.g., articles, workout plans), and what motivates continued use. This includes the effectiveness of personalized content and reminders.

3. Methodology for Usability Testing of Fitness Tracker App

Introduction

The usability testing methodology for the fitness tracker app is designed to comprehensively evaluate the app's user experience, focusing on ease of use, engagement, and overall satisfaction. This methodology encompasses a range of typical tasks for participants to perform, types of data to collect, and recording or measurement methods. The objective is to identify usability issues, understand user behavior, and gather insights to enhance the app's design and functionality.

Typical Tasks for Participants

Task One: Setting Up a Personal Profile and Goals

Objective: task aims to assess the effectiveness of the app's personalization engine in adapting recommendations to the user's evolving interests, engagement levels, and health goals. This encompasses the user's satisfaction with how the app leverages their

interaction data over time to offer more tailored and relevant content, workouts, and mental health advice.

Task Two: Tracking a Workout Session

Participants will be asked to log a workout session using the app. This task includes selecting a workout from provided templates, using the tracking feature during the workout, and reviewing the summary and insights post-workout.

Objective: Evaluate the usability of workout tracking features, including the ease of finding suitable workouts, interacting with tracking functionalities, and understanding post-workout analytics.

Task Three: Accessing and Interacting with Educational Content and Mental Health Resources

In this task, participants will explore the educational articles and mental health section. They will select and read articles of their choice, participate in a guided meditation, and use a mental wellness tool available in the app.

Objective: Determine the effectiveness of the app in providing engaging and useful educational and mental wellness content, and assess how this content contributes to user engagement and satisfaction.

Types of Data to Collect

Quantitative Data:

Task Completion Time: The time taken by participants to complete each task.

Success Rate: The percentage of participants who successfully complete each task without assistance.

Error Rate: The number and types of errors participants encounter during task execution.

Engagement Metrics: Frequency of app usage, time spent on various features, and interaction rates with content and notifications.

Qualitative Data:

User Satisfaction: Participants' subjective ratings of their satisfaction with each task and the app overall.

Feedback on Usability Issues: Open-ended responses detailing difficulties faced, features liked or disliked, and suggestions for improvement.

Perceived Usefulness: Participants' perceptions of the app's value in achieving their fitness and wellness goals.

Recording or Measurement Method and Reason for Choice

Direct Observation and Note-Taking

Method: Researchers will observe participants as they perform each task, taking detailed notes on their actions, reactions, and any verbal comments.

Reason: This method allows for real-time capture of user behavior and immediate feedback, providing insights into user interactions that may not be fully captured through automated metrics.

Screen Recording and Logging

Method: Participants' interactions with the app will be recorded using screen recording software. Additionally, log files will be generated to track all actions within the app.

Reason: Screen recordings and log files offer a comprehensive view of the user's journey through the app, capturing details such as clicks, navigation paths, and time spent on each feature. This data is invaluable for identifying usability issues and understanding user behavior.

Surveys and Interviews

Method: Post-task surveys and follow-up interviews will be conducted to gather qualitative data on user satisfaction, perceived usefulness, and feedback on specific features or experiences.

Reason: Surveys and interviews provide direct insights from users about their experience, allowing for a deeper understanding of their satisfaction levels and the usability of the app. This feedback is crucial for identifying areas for improvement.

Conclusion

This usability testing methodology is designed to offer a comprehensive evaluation of the fitness tracker app, focusing on key user interactions and experiences. By combining tasks that cover the app's primary features with diverse data collection methods, the testing aims to uncover actionable insights that can drive improvements in usability and user satisfaction. Through careful analysis of both quantitative and qualitative data, the development team will be equipped to refine the app, ensuring it meets the needs and expectations of its target users effectively.

4. Informed Consent Form for Usability Testing

An informed consent form is crucial for conducting usability testing ethically and effectively. It ensures participants are fully informed about the study, its purpose, what is expected of them, and how their data will be used. Below is a template for an informed consent form tailored for the usability testing of your fitness tracker app. This template outlines key sections and information typically included in such a document.

Title of Study: Usability Testing of FITBUDDY

Researchers: Arshdeep Ghotra and Nand Bhasin

Introduction

You are being invited to participate in a usability study for Fitbuddy, developed by Arshdeep Ghotra and Nand Bhasin. This study aims to evaluate the app's user interface, features, and overall user experience. Your participation is voluntary, and you may withdraw at any time without penalty.

What Does the Study Involve?

If you agree to participate, you will be asked to:

Complete a series of tasks using the app, such as setting up a profile, tracking a workout, and accessing educational content.

Provide feedback on your experience through surveys and/or interviews.

Allow the session to be recorded for analysis purposes (optional).

The session is expected to last approximately [duration] minutes.

Risks and Benefits

There are minimal risks associated with participating in this study. You may experience typical fatigue from screen time. The benefits include contributing to the improvement of [Fitness Tracker App Name] and potentially enhancing your own user experience with the app.

Confidentiality

Your privacy and the confidentiality of your data are of utmost importance. Any information collected during this study will be anonymized and used solely for research purposes. Recordings and feedback will not be associated with your identity, and any personal information will be stored securely and accessed only by the research team.

Compensation

You'll get a free coffee drink or any meal from your our team

Withdrawal

Your participation in this study is entirely voluntary. You have the right to withdraw at any time without consequence. Should you choose to withdraw, your data will be excluded from the study.

Consent

I have read and understood the information provided about the usability study of Fitbuddy. I have had the opportunity to ask questions and have them answered to my satisfaction. By signing below, I agree to participate in the study under the conditions outlined above.

Participant's Name (Print):

Participant's Signature:

Date: _____

Researcher's Signature (optional):

Date: _____

5. Results of the Usability Testing for Fitness Tracker App

A) Pilot Study

The pilot study for the usability testing of the Fitness Tracker App was conducted over a two-week period in a controlled environment within our tech lab, as well as remotely, to accommodate a diverse group of participants. The selection process aimed at including a wide range of users, from fitness enthusiasts to those new to fitness tracking, to ensure a comprehensive understanding of user experiences. The physical setup included a quiet room with a standard setup of a desk, chair, and provided devices preloaded with the app. For remote participants, guidelines were provided to ensure a controlled testing environment.

Participants were recruited through social media channels, fitness forums, and local community boards, aiming for a sample size of 30 individuals to yield statistically significant insights while managing the

depth of analysis for each session. Testing sessions were scheduled at various times throughout the day to accommodate different user schedules and to assess if the time of day influenced the app's usability or satisfaction.

B) Satisfaction Questions

Participants were asked to rate their satisfaction on a scale of 1 to 5 (1 being very dissatisfied and 5 being very satisfied) based on the following questions:

1. "How satisfied are you with the app's ability to adapt and personalize recommendations (workouts, articles, mental health advice) based on your usage over time?"
2. How satisfied are you with the process of tracking a workout session?
3. How satisfied are you with the variety and relevance of the educational content provided?
4. How satisfied are you with the app's features supporting mental health and meditation?

5. How satisfied are you with the overall user experience and interface of the app?

C) Visualization of Results

The results were visualized using a combination of bar charts and pie charts, highlighting the distribution of satisfaction levels across different aspects of the app. Each question's average satisfaction rating was calculated and presented alongside the percentage distribution of responses.

D) Data Analysis of Each Question

Average Satisfaction Rating: 3.7

Distribution of Responses:

Very Satisfied: 25%

Satisfied: 40%

Neutral: 20%

Dissatisfied: 10%

Very Dissatisfied: 5%

Workout Session Tracking:

Average satisfaction rating: 3.8

Feedback indicated a need for more detailed tracking options and a clearer summary of workout analytics.

Educational Content:

Average satisfaction rating: 4.0

Users appreciated the variety but desired more depth and personalization in the content.

Mental Health and Meditation:

Average satisfaction rating: 3.5

The feedback pointed towards a need for more guided meditation options and interactive mental health tools.

Overall User Experience and Interface:

Average satisfaction rating: 4.1

While the app was generally praised for its clean design, some users noted occasional navigational difficulties.

E) List of Things That Need to Be Improved

Based on the pilot study's findings, several areas require improvement:

Enhance Workout Tracking: Incorporate more granular tracking features and provide a more comprehensive summary post-workout to increase user satisfaction.

Deepen Educational Content: Expand the educational articles and videos to cover more topics and include advanced insights, tailoring content to user preferences and goals.

Expand Mental Health Resources: Introduce a wider variety of guided meditations and interactive tools to support mental well-being, catering to different user needs.

Refine User Interface: Address navigational issues highlighted by users to ensure a smoother, more intuitive experience across the app.

Personalization Features: Increase the app's ability to personalize content, workouts, and health tips based on user behaviour and preferences to enhance user engagement and satisfaction.